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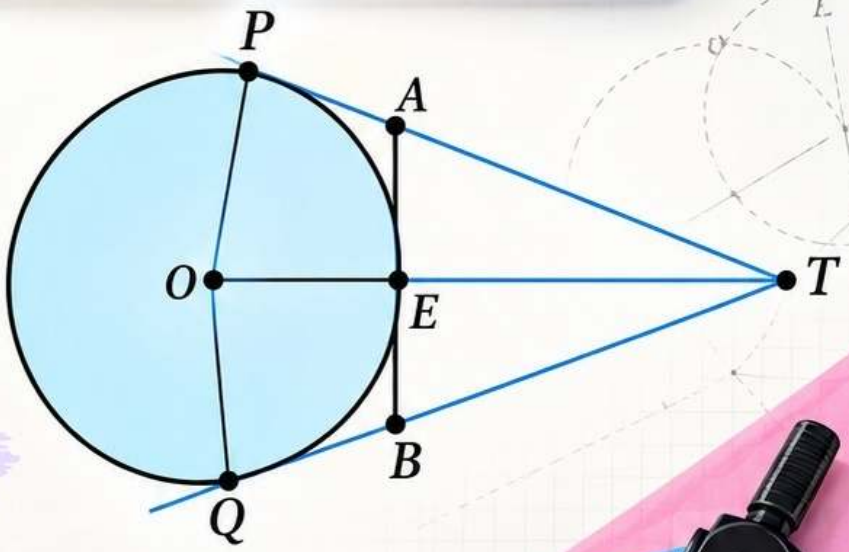
CLASS X MATHEMATICS

CHAPTER – 10

CIRCLES

SOLVED & UNSOLVED IMPORTANT QUESTIONS

- 10 Solved Examples
- 10 Unsolved Examples
- Exam Oriented
- Based on Latest CBSE Pattern



Circles

- Tangents
- Chords
- Angles in a Circle
- Concyclic Points

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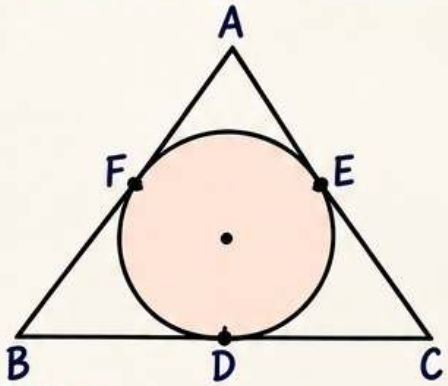
Ex.1 & 2

CLASS X MATHS

CHAPTER - 10 : CIRCLES

IMPORTANT QUESTIONS WITH SOLUTIONS

Ex. 1 In the given figure, the incircle of ΔABC touches the sides BC, CA and AB at D, E, F respectively.



Prove that :

$$AF + BD + CE = AE + CD + BF = \frac{1}{2}(\text{perimeter of } \Delta ABC)$$

Solution :

We know that the lengths of tangents from an exterior point to a circle are equal.

$\therefore$ $AF = AE$ ..... (i) [tangents from A]
$BD = BF$ ..... (ii) [tangents from B]
$CE = CD$ ..... (iii) [tangents from C]

Adding (i), (ii) and (iii), we get

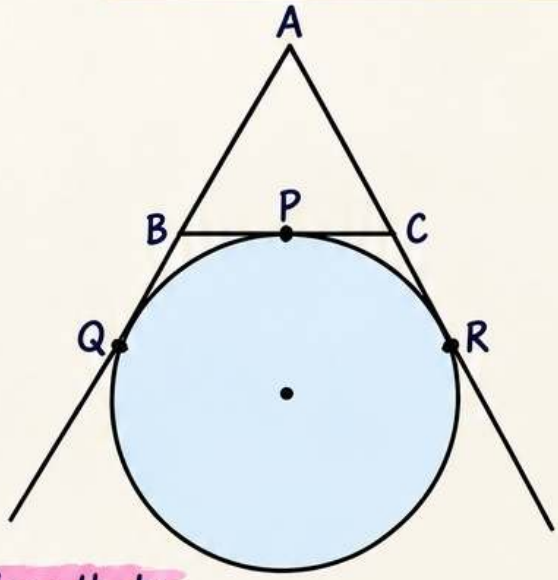
$(AF + BD + CE) = (AE + BF + CD) = k$ (say)

$$\text{Perimeter of } \Delta ABC = (AF + BD + CE) + (AE + BF + CD) = (k + k) = 2k$$

$$\therefore \quad k = \frac{1}{2}(\text{perimeter of } \Delta ABC)$$

Hence, $AF + BD + CE = AE + CD + BF = \frac{1}{2}(\text{perimeter of } \Delta ABC)$.

Ex. 2 A circle touches the side BC of a ΔABC at P, and touches AB and AC produced at Q and R respectively, as shown in the figure.



Show that :

$$AQ = \frac{1}{2}(\text{perimeter of } \Delta ABC)$$

Solution :

We know that the lengths of tangents from an exterior point to a circle are equal.

$\therefore$ $BQ = BP$ ..... (i) [tangents from B]
$CR = CP$ ..... (ii) [tangents from C]
$AQ = AR$ ..... (iii) [tangents from A]

Perimeter of ΔABC
$= AB + BC + AC$
$= AB + BP + CP + AC$
$= AB + BQ + CR + AC$ [using (i) and (iii)]
$= AQ + AR$
$= 2AQ$ [using (iii)]

Hence, $AQ = \frac{1}{2}(\text{perimeter of } \Delta ABC)$.

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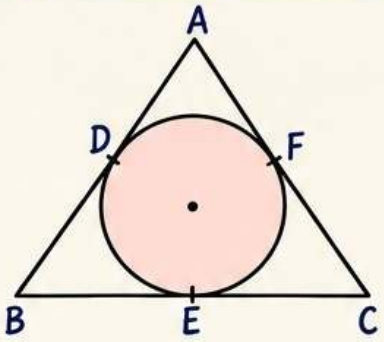
CLASS X MATHS

CHAPTER – 10 : CIRCLES

Ex.3, 4 & 5

IMPORTANT EXAMPLES WITH SOLUTIONS

Ex. 3 In Fig., if AB = AC, prove that BE = EC.



Sol.

Since tangents from an exterior point to a circle are equal in length.

$\therefore$ AD = AF [Tangents from A]
BD = BE [Tangents from B]
CE = CF [Tangents from C]

Now, AB = AC

$\Rightarrow$ AB – AD = AC – AD [Subtracting AD from both sides]

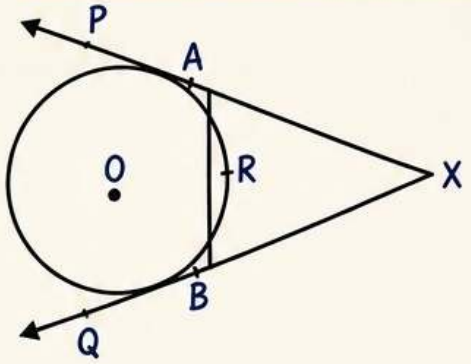
$\Rightarrow$ AB – AD = AC – AF [Using (i)]

$\Rightarrow$ BD = CF $\Rightarrow$ BE = CF [Using (iii)]

$\Rightarrow$ BE = CE [Using (iii)]

Hence, BE = EC.

Ex. 4 In fig. XP and XQ are tangents from X to the circle with centre O. R is a point on the circle. Prove that XA + AR = XB + BR.



Sol.

Since lengths of tangents from an exterior point to a circle are equal.

$\therefore$ XP = XQ ..... (i) [From X]
AP = AR ..... (ii) [From A]
BQ = BR ..... (iii) [From B]

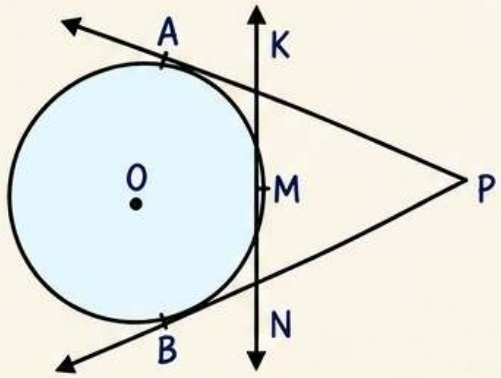
Now, XP – XQ

$\Rightarrow$ XA + AP = XB + BQ

$\Rightarrow$ XA + AR = XB + BR [Using (ii) and (iii)]

Hence, XA + AR = XB + BR.

Ex. 5 PA and PB are tangents from P to the circle with centre O. At point M, a tangent is drawn cutting PA at K and PB at N. Prove that KN = AK + BN.



Sol.

We know that the tangents drawn from an external point to a circle are equal in length.

$\therefore$ PA = PB ..... (i) [From P]
KA = KM ..... (ii) [From K]
NB = NM ..... (iii) [From N]

Adding equations (ii) and (iii), we get

KA + NB = KM + NM

$\Rightarrow$ AK + BN = KM + MN = KN

Hence, KN = AK + BN.

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CHAPTER - 10 : CIRCLES

IMPORTANT EXAMPLES WITH SOLUTIONS

Ex.6 & 7

Ex. 6

ABC is a right triangle, right angled at B. A circle is inscribed in it. The lengths of the two sides containing the right angle are 6 cm and 8 cm. Find the radius of the incircle.

Sol.

Let the radius of the incircle be x cm.

In the given figure, the incircle touches the side AB, BC and CA at D, E, F respectively.

Let O be the centre of the circle.

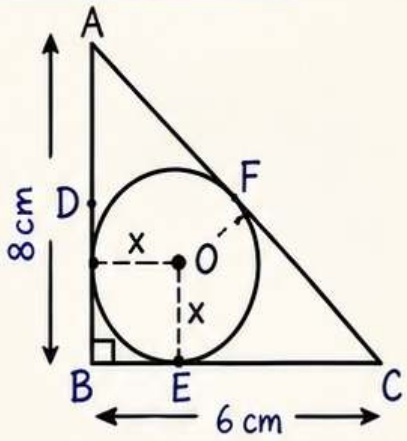
Then, OD = OE = OF = x cm.

Also, AB = 8 cm and BC = 6 cm.

Since the tangents to a circle from an external point are equal, we have

$$AF = AD = (8 - x) \text{ cm, and}$$
$$CF = CE = (6 - x) \text{ cm.}$$

$\therefore$ $AC = AF + CF = (8 - x) + (6 - x)$ cm
$= (14 - 2x)$ cm.



Now, $AC^2 = AB^2 + BC^2$

$\Rightarrow$ $(14 - 2x)^2 = 8^2 + 6^2$

$\Rightarrow$ $(14 - 2x)^2 = 100$

$\Rightarrow$ $14 - 2x = \pm 10$

$\Rightarrow$ $x = 2$ or $x = 12$

$\Rightarrow$ $x = 2$ [neglecting $x = 12$].

Hence, the radius of the incircle is 2 cm.

Ex. 7

In fig., circles C(O, r) and C(O', r/2) touch internally at a point A and AB is a chord of the circle C (O, r) intersecting C (O', r/2) at C. Prove that AC = CB.

Sol.

Join OA, OC and OB. Clearly, $\angle OCA$ is the angle in a semi-circle.

$$\therefore \angle OCA = 90^\circ$$

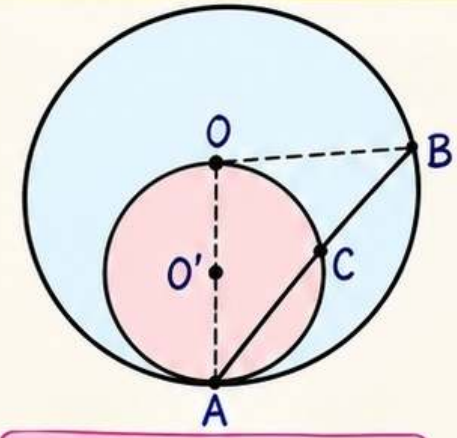
In right triangles OCA and OCB, we have

$$OA = OB = r$$
$$\angle OCA = \angle OCB = 90^\circ$$
$$\text{and } OC = OC$$

So, by RHS criterion of congruence, we get

$\Delta OCA \cong \Delta OCB$

$\Rightarrow$ AC = CB



Hence, AC = CB.

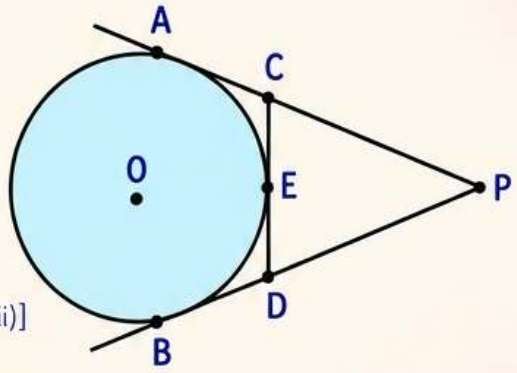
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CHAPTER - 10 : CIRCLES
IMPORTANT QUESTIONS WITH SOLUTIONS
Ex. 8, 9 & 10

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Ex. 8

From an external point P, tangents PA and PB are drawn to a circle with centre O. If CD is the tangent to the circle at a point E and PA = 14 cm, find the perimeter of △PCD.

Sol.



From an external point, the tangents to a circle are equal.

$\therefore$ PA = PB = 14 cm [Tangents from P] ... (i)

From point C, the tangents CA and CE are equal.

$\therefore$ CA = CE [Tangents from C] ... (ii)

From point D, the tangents DB and DE are equal.

$\therefore$ DB = DE [Tangents from D] ... (iii)

Now, perimeter of △PCD = PC + CD + PD

= (PC) + (CE + DE) + (PD) [Using (ii) and (iii)]

= (PC + CE) + (PD + DE)

= (PC + CA) + (PD + DB) [Using (ii) and (iii)]

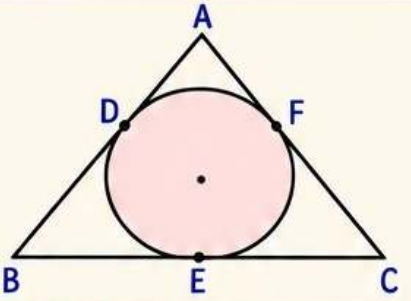
= PA + PB = 14 + 14 = 28 cm [Using (i)]

Hence, the perimeter of △PCD = 28 cm.

Ex. 9

A circle is inscribed in a △ABC having AB = 10 cm, BC = 12 cm and CA = 8 cm and touching these sides at D, E, F respectively, as shown in the figure. Find AD, BE and CF.

Sol.



Let AD = AF = x, BD = BE = y and CE = CF = z.

Given: AB = AD + DB = x + y = 10 ... (i)

BC = BE + EC = y + z = 12 ... (ii)

CA = AF + FC = x + z = 8 ... (iii)

From (i), $y = 10 - x \Rightarrow (10 - x) + z = 12 \Rightarrow z = 2 + x$

From (iii), $x + z = 8 \Rightarrow x + (2 + x) = 8 \Rightarrow 2x = 6 \Rightarrow x = 3$

Then, $y = 10 - x = 10 - 3 = 7$ and $z = 2 + x = 2 + 3 = 5$

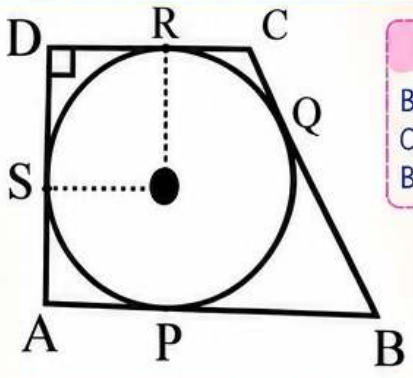
Hence, AD = 3 cm, BE = 7 cm and CF = 5 cm.

Given :
AB = 10 cm BC = 12 cm
CA = 8 cm

Ex. 10

In the given figure, ABCD is a quadrilateral in which ∠D = 90°. A circle C(O, r) touches the sides AB, BC, CD and DA at P, Q, R, S respectively. If BC = 38 cm, CD = 25 cm and BP = 27 cm, find the value of r.

Sol.



Since tangents from an external point to a circle are equal.

From B : BP = BQ = 27 cm ... (i)

$\therefore$ QC = BC − BQ = 38 − 27 = 11 cm ... (ii)

From C : CQ = CR = 11 cm ... (iii)

$\therefore$ DR = CD − CR = 25 − 11 = 14 cm ... (iv)

In right angle at D, the tangents from D are equal.

$\therefore$ DR = DS = 14 cm ... (v)

But the radius of the circle is DS = r.

Hence, r = 14 cm.

Given :
BC = 38 cm
CD = 25 cm
BP = 27 cm

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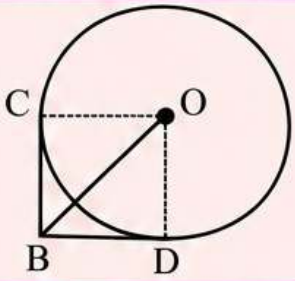
CLASS X MATHS

CHAPTER - 10 : CIRCLES

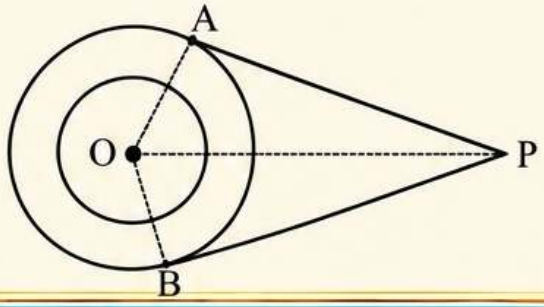
Que. 1 to 5

UNSOLVED QUESTIONS FOR PRACTICE

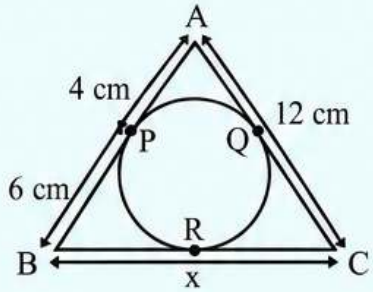
Q.1 Two tangent segments BC and BD are drawn to a circle with centre O such that $\angle CBD = 120°$. Prove that $OB = 2BC$.



Q.2 In the given figure, O is the centre of two concentric circles of radii 4 cm and 6 cm respectively. PA and PB are tangents to the outer and inner circle respectively. If PA = 10 cm, find the length of PB up to one place of decimal.

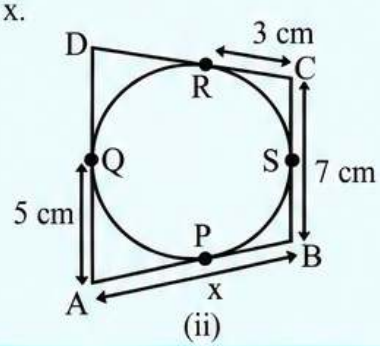


Q.3 (a) In the figure (i) given below, triangle ABC is circumscribed, find x.



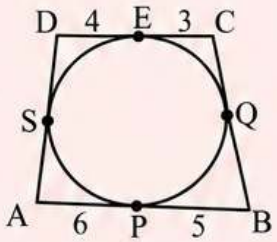
(i)

(b) In the figure (ii) given below, quadrilateral ABCD is circumscribed, find x.



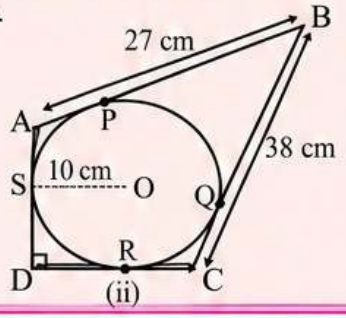
(ii)

Q.4 (a) In the figure (i) given below, quadrilateral ABCD is circumscribed; find the perimeter of quadrilateral ABCD.



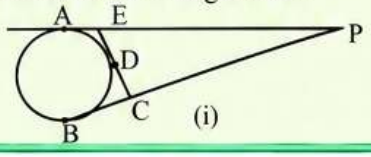
(i)

(b) In the figure (ii) given below, quadrilateral ABCD is circumscribed and $AD \perp DC$, find x if radius of incircle is 10 cm.



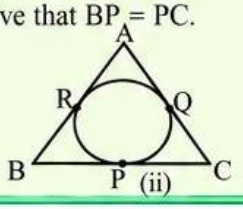
(ii)

Q.5 (a) In the figure (i) given below, from an external point P, tangents PA and PB are drawn to a circle. CE is tangent to the circle at D. If AP = 15 cm, find the perimeter of the triangle PEC.



(i)

(b) In the figure (ii) given below, the incircle of ΔABC touches the sides BC, CA and AB at points P, Q and R respectively. If AB = AC, prove that BP = PC.

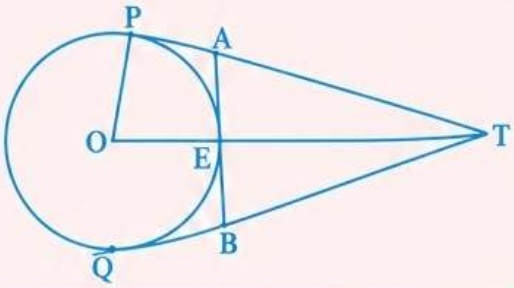


(ii)

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Q.6

In Fig. 9.20. O is the centre of a circle of radius 5 cm, T is a point such that OT = 13 cm and OT intersects the circle at E. If AB is the tangent to the circle at E, find the length of AB.



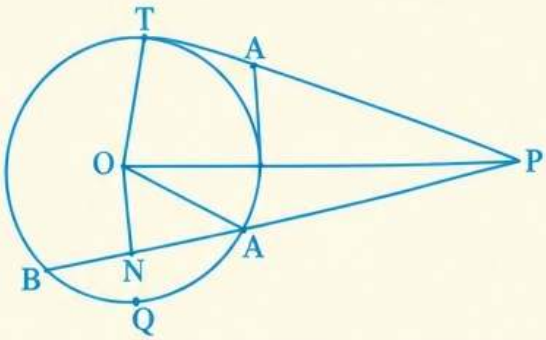
Q.7

Sample Question 1 : In Fig. 9.15, from an external point P, a tangent PT and a line segment PAB is drawn to a circle with centre O. ON is perpendicular on the chord AB. Prove that :

(i) $PA \cdot PB = PN^2 - AN^2$

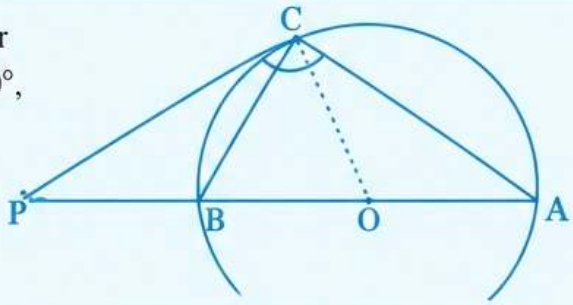
(ii) $PN^2 - AN^2 = OP^2 - OT^2$

(iii) $PA \cdot PB = PT^2$



Q.8

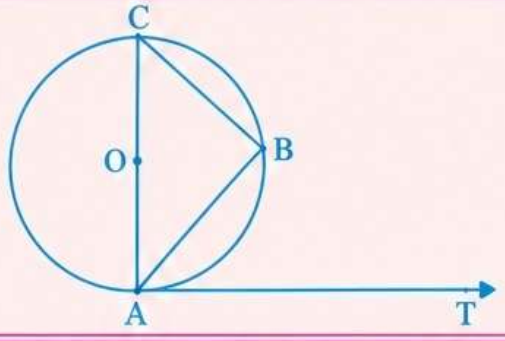
The tangent at a point C of a circle and a diameter AB when extended intersect at P. If $\angle PCA = 110°$, find $\angle CBA$ [see Fig. 9.21].



Q.9

If AB is a chord of a circle with centre O, AOC is a diameter and AT is the tangent at A as shown in Fig. 9.17. Prove that

$\angle BAT = \angle ACB$



Q.10

In Fig. 9.18, tangents PQ and PR are drawn to a circle such that $\angle RPQ = 30°$. A chord RS is drawn parallel to the tangent PQ. Find the $\angle RQS$.

[**Hint:** Draw a line through Q and perpendicular to QP.]

